Chapter II

Sequences

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1 Introduction and Motivation

In the opening chapter some problems involving infinite series were discussed. It was seen that treating infinite series as if they were truly infinite summations led to contradictions. However, it was found that in some cases treating infinite series as infinite summations was not only allowed, but also incredibly useful. This was seen in the example of integrating e^{-x^2}. This then begged the question, why can some infinite series be treated as infinite summations and others not? Further, which infinite series can be integrated, or even differentiated term by term, and which cannot? These questions seem difficult at the moment and we must first build up an understanding of some properties of infinite series in order to have a chance of answering them. What we can say though is that it seems that the necessary condition for an infinite series to have any hope of satisfying the usual properties of a summation is that it *converges*. So far, we have avoided defining what convergence means in a formal capacity. This very much mirrors the historical development of convergent infinite series in which a formal definition of convergence was not developed until mathematicians reached the impasse that was Fourier series. Intuitively, we would like to say that a series converges to some value, say x, if as we add more terms to the summation, the closer it gets to the value x. For example, consider the finite summation

$$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}=1.96875.$$

In this case, the more terms that are added, the closer and closer the sum gets to the number 2. With this, it is incredibly tempting to simply write

$$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\cdots=2$$

and whilst this can lead to some useful results, inevitably leads to contradictions. Instead, it is better to think of the value of the series for some finite number of terms. This way at each step, the series really is just some finite summation and therefore satisfies the usual properties for addition. For example, summing one term gives

$$a_1=1.$$

Summing two terms gives

$$a_2=1+\frac{1}{2}=\frac{3}{2}$$

and summing three terms gives

$$a_3=1+\frac{1}{2}+\frac{1}{4}=\frac{7}{4}.$$

At each step, we are defining a term in a *sequence*. In this case, this sequence is given by

$$\left(1, \frac{3}{2}, \frac{7}{4}, \frac{15}{8}, \frac{31}{16}, \dots\right).$$

The question of asking whether or not the infinite series *converges* is now recast into asking whether or not the sequence *converges*. It turns out that by recasting infinite series as sequences, where each term of the sequence is defined by a finite summation, the question of convergence of infinite series becomes much easier to tackle. We will therefore spend a serious amount of time building up a framework for analysing the nature of sequences, and in particular what it means for them to converge.

2 Sequences

As outlined above, in order to develop a framework for studying infinite series we first require a clear understanding of the theory of sequences. In fact, we will see that that a large majority of concepts we study in this course can be reduced to statements about the behaviour of sequences, including the completeness axiom.

A sequence can be simply understood as an ordered list of numbers with no end. For example,

$$(1, 4, 9, 16, \dots) \tag{2.1}$$

is a sequence and

$$(\pi, 3, 5, e, \dots) \tag{2.2}$$

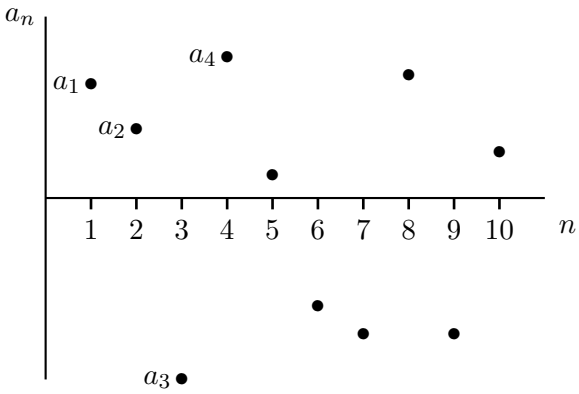
is also a sequence. Note that these are different from sets since the order of a sequence matters.

Rather than inventing some new formal term for a sequence, it is fruitful to realise that they are nothing more than functions from the natural numbers to the real numbers. That is, you can think of the sequence in (2.1) as a mapping from 1 to 2, 2 to 4, 3 to 9 and so on. Formally, the definition is then given by the following.

Definition 2.1: Sequence

A ***sequence*** is a function $a : \mathbb{N} \to \mathbb{R}$.

Whilst this may initially seem more convoluted to think of sequences as functions, it is actually much simpler this way since we already have a fleshed out understanding of functions. Further, since functions can be represented graphically, we can also represent sequences graphically as follows.



Here we have labelled the axes and the first 4 points, but we won't usually do so. Many of the proofs of the course will be inspired by first thinking about the graphical picture. That isn't to say any proofs will rely on pictures, but a substantial amount of intuition can be drawn from them.

Remark 2.2: Sequence Notation

(i) For a function $a : \mathbb{N} \to \mathbb{R}$ you would normally denote it by $a(n)$. For example, (2.1) would be written as

$$a(1) = 1, \quad a(2) = 4, \quad a(3) = 9, \quad a(4) = 16, \quad \dots \quad .$$

However, it is the convention that the subscript notation is used to denote the terms of a sequence. That is

$$a_1 = 1, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 16, \quad \dots \quad .$$

(ii) In some special cases, the terms a a given sequence can be written explicitly. For example, (2.1) can simply be written as

$$a_n = n^2$$

whilst the sequence given in (2.2) cannot be written this way since there is no clear explicit formula for this sequence. The term a_n is referred to as the *n-th term of the sequence.*

(iii) Some sequences can be defined recursively. For example, the sequence given by the expression

$$a_n = a_{n-1} + a_{n-2}$$

with $a_1 = 1$ and $a_2 = 1$. This defines the *Fibonacci sequence* and is given explicitly by

$$(1, 1, 2, 3, 5, \dots).$$

(iv) In general, a sequence is denoted by (a_n) but can also be denoted by $(a_1, a_2, a_3, \dots)$ or $(a_n)_{n=1}^{\infty}$. In this text we will mostly use the first two notations.

We will round off this discussion of sequences with a few quick fire examples.

Example 2.3: Sequences

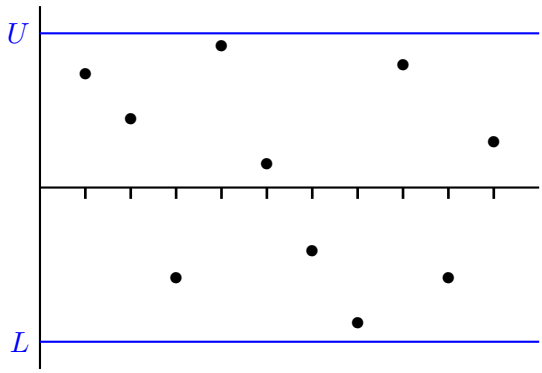
(i) $\left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right)$.

(ii) (a_n) with $a_n = n^3$.

(iii) (b_n) with $b_{n+1} = \frac{b_n}{2}$ and $b_1 = 1$.

3 Bounded Sequences

A key property of a sequence is that it is never ending. It is the goal of this section to characterise the behaviour of sequences at their 'tail' end. Like with most areas of mathematics, it is useful to restrict our study to a class of simpler objects and build from there. In this case, it will be useful to start by considering sequences that are controlled or constrained to a specific region. What would a constrained, or bounded, sequence look like pictorially? A rough sketch of this could be as follows.



Here, U, L are simply real numbers. Now, as discussed before, pictures will not be sufficient for rigorous proofs nor concrete definitions. They are useful however as a stepping stone to producing concrete definitions. When recalling definitions, or even constructing proofs of theorems, it can be useful to keep a picture in mind and build from there. With this picture in mind, we then arrive at the definition of a bounded sequence.

Definition 3.1: Bounded Sequence

Let (a_n) be a sequence. Then (a_n) is said to be ***bounded above*** if there exists a $U \in \mathbb{R}$ such that $a_n \leq U$ for all $n \in \mathbb{N}$. The number $U \in \mathbb{R}$ is called a ***upper bound*** for (a_n).

Similarly, a sequence (a_n) is said to be ***bounded below*** if there exists a $L \in \mathbb{R}$ such that $L \leq a_n$ for all $n \in \mathbb{N}$. The number $L \in \mathbb{R}$ is called a ***lower bound*** for (a_n).

Finally, a sequence (a_n) is said to be ***bounded*** if it is bounded from both above and below. If a sequence is not bounded, it is said to be ***unbounded.***

The final definition of a bounded sequence can be written more succinctly via the following theorem.

Theorem 3.2: Bounded Sequence

A sequence (a_n) is bounded if and only if there exists a $K \in \mathbb{R}$ such that $|a_n| \leq K$ for all $n \in \mathbb{N}$.

Proof. Exercise □

With the definition and theorem we can new begin to prove some properties of sequences. Namely, if they are bounded or not.

Example 3.3: Bounded/Unbounded Sequences

(i) Let (a_n) be a sequence given by $a_n = n$. Then (a_n) sequence is bounded below.

Proof. Since $n > 0$ for all $n \in \mathbb{N}$ it follows (a_n) is bounded below by 0. □

(ii) Let (b_n) be a sequence given by $b_n = 1 + \sin(n)$. Then (b_n) is bounded.

Proof. Since $|\sin(n)| \leq 1$ for all $n \in \mathbb{N}$, it follows that

$$|b_n| = |1 + \sin(n)| \leq 1 + |\sin(n)| \leq 2.$$

Hence (b_n) is bounded. □

(iii) Let (c_n) be the sequence given by $c_n = (-1)^n n$. Then (c_n) is unbounded.

Proof. Assume towards a contradiction that (c_n) is bounded. That is, there exists a $K \in \mathbb{R}$ such that

$$|(-1)^n n| \leq K$$

for all $n \in \mathbb{N}$. By properties of the absolute value function, and the fact that $n > 0$ for all $n \in \mathbb{N}$, this is equivalent to

$$n \leq K$$

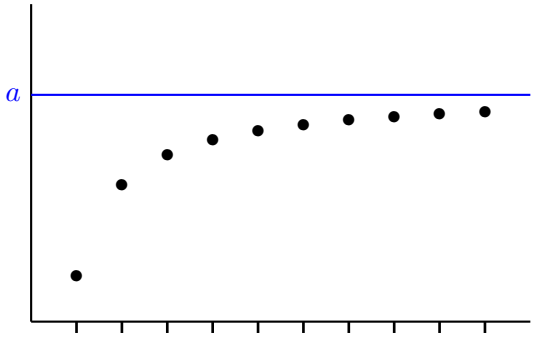
for all $n \in \mathbb{N}$. This is in direct contradiction with the Archimedean property. Hence (c_n) is unbounded. □

That is the end of the section on bounded sequences. We will see that by being able to identify if a sequence is bounded will be crucial for probing the long term

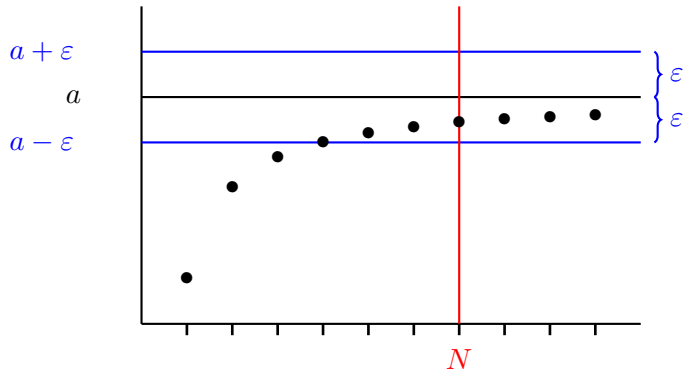
nature of sequences.

4 Convergent Sequences

We now arrive at perhaps one of the most important sections in this entire course, namely the topic of convergent sequences. You may have even come across this term in previous courses and have an intuitive notion as to what it means for a sequence to converge. One point we have emphasised though is that intuition is not sufficient enough foundation to build on. Indeed, it can get you quite far with some calculations, but eventually you will always meet an impasse. Saying this, like the notion of a bounded sequence, intuition will play an important roll in producing a rigorous definition. So what is this intuitive notion of convergence of a sequence? Alternatively, what would the picture of a convergent sequence look like? The following picture gives an example of this.



So here, it appears as if the sequence is convergent to some value $a \in \mathbb{R}$ if it gets arbitrarily close to a as we continue to the right. It might be incredibly tempting to simply say now that the sequence reaches the point a at infinity. However, this is exactly what leads to the contradictions that were outlined in the opening chapter. The key point is that we need to make a distinction between the sequence getting arbitrarily close to the value a and it equalling it. How can we capture this notion in a precise and rigorous definition? Moving in this direction, we realise that an equivalent way of realising a sequence is convergent if that the sequence is eventually constrained to some arbitrarily small, but never vanishing, region around a. We can visualise this via the following picture.



Here, the value of $\varepsilon > 0$ controls the size of the region about a. The closer ε is to zero, the tighter the region around a becomes. Note that it is specifically chosen

so that $\varepsilon \neq 0$ and this ensures the region around a never completely vanishes. This is the first step to our definition of convergence, namely defining a region around a point of arbitrary size, but that never vanishes. This leads us to our first definition.

Definition 4.1: ε-Neighbourhood

Let $\varepsilon > 0$. Then the ***ε-neighbourhood*** $V_\varepsilon(a)$ of a point $a \in \mathbb{R}$ is the interval

$$V_\varepsilon(a) := (a - \varepsilon, a + \varepsilon) = \{x \in \mathbb{R} : a - \varepsilon < x < a + \varepsilon\}.$$

With the definition of ε-neighbourhood in hand, we are one step closer to defining convergent sequences. Recall that our intuitive notion of convergence led us to the idea that a sequence is convergent if that sequence is eventually constrained to some arbitrarily small, but never vanishing, region around a. We have our rigorous definition of an arbitrarily small, but never vanishing, region around a but we now need to formalise the notion of the sequence eventually being constrained to that region. This is quite simple. All we need to say is that that there is some point in the sequence for which any point after it is contained in the region. Piecing all these components together gives us our formal definition for a convergent sequence.

Definition 4.2: Topological Convergence

A sequence (a_n) ***converges*** to $a \in \mathbb{R}$ if, given any ε-neighbourhood of a, there exists a point in the sequence after which all the of terms are in $V_\varepsilon(a)$.

Note that the word *any* in the above definition is capturing the notion *arbitrary size* of the region around a that was mentioned above. What is hopefully clear from the above definition, and the pictures above, is that if the sequence is convergent, the point in the sequence for which all points after it are contained in the region is dependent on ε. In general, the closer ε is to zero, the further one must travel along the sequence until it is eventually constrained to the region. Finding this value of the sequence will be a key component to proving the convergence of sequences.
Whilst the above definition is perfectly valid and gives no contradictions, it is not the most useful for proving the convergence of sequences explicitly. For this, we simply unravel the definition above so as to be more explicit with what needs to be proven. This leads us nicely to the equivalent definition of convergence.

Definition 4.3: Metric Convergence

A sequence (a_n) ***converges*** to $a \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n > N$.

It should hopefully be clear how these definitions are equivalent. The ε-neighbourhood has simply been unravelled by using the equivalence

$$a - \varepsilon < x < a + \varepsilon \Leftrightarrow |x - a| < \varepsilon$$

and the phrase 'there exists a point in the sequence after' has now be written as 'there exists an $N \in \mathbb{N}$' and 'for all $n > N$'.

This may initially seem superfluous, but this definition makes it much clearer in what must be proven in order to show that a sequence is convergent. Additionally, it will be useful to prove further more general statements about convergent sequences. Before proving the convergence of some sequences, it is fruitful to introduce the barrage of notation that can sometimes be see for convergent sequences.

Remark 4.4: Sequence Notation

The following notations are all equivalent for a convergent sequence (a_n) to converge to a

(i) $\lim(a_n) = a$.

(ii) $a_n \to a$ as $n \to \infty$.

(iii) $a_n \to a$.

(iv) $\lim_{n\to\infty}(a_n) = a$.

We will mostly adopt the first notation, but other texts my use other notations.

Finally, with all these definitions and notations to hand, we are able to prove the convergence, or not, of sequences.

Example 4.5: Convergent Sequences

Let (a_n) be the sequence given by $a_n = 1/n$. Then $\lim(a_n) = 0$.

Proof. By the Archimedean property, it follows that for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\frac{1}{N} < \varepsilon.$$

Hence, for every $n > N$ it follows that

$$\begin{aligned} |a_n - 0| &= \left|\frac{1}{n}\right| \\ &< \varepsilon. \end{aligned}$$

Thus $\lim(a_n) = 0$. □

Since sequences are have their domain in $\mathbb{N}$, you will often find that the existence of such an $N \in \mathbb{N}$ depends on the Archimedean property. The above example is slightly deceptive in that the proving of the convergence of a sequence rarely takes this clean simple form. In particular, the choice of N seems to appear out of nowhere in Example 4.5. In fact, the majority of the work done to produce these proofs is in finding this value of N. This preliminary work is referred to as *scratch work* and is often omitted in many textbooks for the sake of brevity. It can often give the illusion that certain proofs come from nowhere and are simply beamed down by the mathematical deities. The reality is far from this, and behind most of these proofs lies piles and piles of crumpled paper of failed attempts from the authors. This is often why the offices of mathematicians are littered with piles

of paper. These piles do not contain clean, beautiful and short proofs. On the contrary, chances are they contain nonsense.

Example 4.6: Convergent Sequences

Let (a_n) be the sequence given by $a_n = \dfrac{3n+1}{n+2}$. Then $\lim(a_n) = 3$.

Scratch Work. We want to construct an inequality of the form

$$|a_n - 3| < \frac{k}{n}$$

for some $k \in \mathbb{R}$. If we can do this we can use the Archimedean property to find an $N \in \mathbb{N}$ so that $k/n < \varepsilon$ for all $n > N$. To this end, let's start manipulating $|a_n - 3|$ using all the properties of the absolute value function at our disposal. This gives

$$\begin{aligned} |a_n - 3| &= \left| \frac{3n+1}{n+2} - 3 \right| \\ &= \left| \frac{3n+1-3(n+2)}{n+2} \right| \\ &= \left| \frac{-5}{n+2} \right| \\ &= \frac{5}{n+2} \\ &< \frac{5}{n} \end{aligned}$$

We can now construct our proof.

Proof. By the Archimedean property, it follows that for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\frac{5}{N} < \varepsilon.$$

Hence, for every $n > N$ it follows that

$$|a_n - 3| < \frac{5}{n} < \varepsilon$$

□

As you probably realise, the above proof of the example above is more complicated than the previous one. However, this example follows the more familiar pattern for how to prove the convergence of certain sequences. This isn't to say that proofs of this type follow some rigid algorithm similar to say how you may find the solutions to a quadratic equation. But, there is a rough pattern that explicit convergence proofs follow that we can extrapolate a procedure from. This procedure is only a guide though, and moving from one step to the next can vary greatly in difficulty depending on the specific sequence.

Remark 4.7: Proving Convergence of Sequences

To prove that a sequence (a_n) converges to a, we can use the following rough procedure:

1. Expand the expression $|a_n - a|$.
2. From here, chain together a series of inequalities until you arrive at $|a_n - a| < k/n$ for some $k \in \mathbb{R}$. It is useful to keep in mind the properties of the absolute value function to produce this chain of inequalities.
3. By the Archimedean property, for every $\varepsilon > 0$ let $N \in \mathbb{N}$ such that $k/N < \varepsilon$.
4. Use this to prove that $\lim(a_n) = a$.

To demonstrate this procedure, and further possible complications that can arise in moving between the steps, consider this final example.

Example 4.8: Convergent Sequences

Let (a_n) be the sequence given by $a_n = \dfrac{3n^2 + 2n - 10}{n^2 - 3n - 3}$. Then $\lim(a_n) = 3$.

Scratch Work. Following the pattern outlined above, we have

$$\begin{aligned} |a_n - 3| &= \left| \frac{3n^2 + 2n - 10}{n^2 - 3n - 3} - 3 \right| \\ &= \left| \frac{3n^2 + 2n - 10 - 3(n^2 - 3n - 3)}{n^2 - 3n - 3} \right| \\ &= \left| \frac{11n - 1}{n^2 - 3n - 3} \right| \\ &< \frac{11n}{|n^2 - 3n - 3|}. \end{aligned}$$

In order to get an inequality less that k/n for some $k \in \mathbb{R}$, we wish to find some $N \in \mathbb{N}$ such that $\frac{1}{2}n^2 < n^2 - 3n - 3$ for all $n > N$. This is true when

$$n^2 - 6n - 6 > 0$$

which in turn is true if $n \geq 7$. Hence,

$$|a_n - 3| < \frac{11n}{n^2/2} = \frac{22}{n}$$

for $n > 6$. With this, we can now construct our proof.

Proof. By the Archimedean property, for every $\varepsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that $\dfrac{22}{N_1} < \varepsilon$. Let $N = \max(7, N_1)$. It follows that for every $n > N$, we have

$$|a_n - 3| < \frac{22}{n} < \varepsilon.$$

Hence, $\lim(a_n) = 3$. □

Each of these examples have a similar end point, but the journey to get there is drastically different. Once again, this fundamentally comes down to the fact that we must chain together a series of inequalities and those inequalities must be imaginatively produced so as to get the desired result. The problem is that there are infinitely many inequalities that can be satisfied in these examples and most of them are useless. We must always keep in mind the end goal so as to produce this useful chain of inequalities.

An additional thing to note in each of these examples is that the limit was given from the outset. Obviously, this is not normally the case and we will need to construct additionally machinery in order to calculate these explicitly. Additionally, we currently have no way of telling if a single convergent sequence could converge to two different values. This would be problematic and appear to conflict with our intuitive notions of convergence. Thankfully, this is not the case and is proven in the following theorem.

Theorem 4.9: Uniqueness of Limits

The limit of a convergent sequence is unique.

Proof. Let (a_n) be a convergent sequence and assume towards a contradiction that $\lim(a_n) = a$ and $\lim(a_n) = \tilde{a}$ with $a \neq \tilde{a}$. Then for every $\varepsilon > 0$, there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for all $n > N_1$. Similarly, there exists an $N_2 \in \mathbb{N}$ such that

$$|a_n - \tilde{a}| < \varepsilon$$

for all $n > N_2$. Define $N = \max(N_1, N_2)$. It follows that

$$\begin{aligned} |\tilde{a} - a| &= |a_n - a - (a_n - \tilde{a})| \\ &\leq |a_n - a| + |a_n - \tilde{a}| \\ &< 2\varepsilon, \end{aligned}$$

for each $n > N$. Since this is true for every $\varepsilon > 0$, it must be true for $\varepsilon = \frac{1}{3}|\tilde{a} - a|$. Combining this with the above inequality gives

$$\begin{aligned} |\tilde{a} - a| &< \frac{2}{3}|\tilde{a} - a| \\ \Rightarrow 1 &< \frac{2}{3}. \end{aligned}$$

This is a contradiction, hence $a = \tilde{a}$. □

5 Divergent Sequences

As you might expect, not all sequences converge. Sequences that do not converge are aptly known as *divergent sequences*. However, unlike convergent sequences, they can have distinctly different properties. These properties are outlined in the following definition.

Definition 5.1: Divergent Sequences

A sequence (a_n) is said to ***diverge*** if it does not converge. Divergent sequences can the be characterised into the follows three forms:

(i) (a_n) ***diverges to*** ∞ if for all $M > 0$, there exists an $N \in \mathbb{N}$ such that $a_n > M$ for every $n > N$.

(ii) (a_n) ***diverges to*** $-\infty$ if for all $M < 0$, there exists an $N \in \mathbb{N}$ such that $a_n < M$ for every $n > N$.

(iii) Otherwise, (a_n)'s limit ***does not exist***.

It is worth mentioning that if a sequence diverges to ∞ or $-\infty$ it is most definitely *not convergent*. We would never say that '(a_n) converges to ∞'. Saying this, it is common for the following abuse of notation to be used.

Remark 5.2: Divergent Sequences

The following notation is sometimes used for divergent sequences that diverge to ∞ or $-\infty$.

(i) $\lim(a_n) = \infty$ or $\lim(a_n) = -\infty$.

(ii) $a_n \to \infty$ or $a_n \to -\infty$.

(iii) $\lim_{n\to\infty}(a_n) = \infty$ or $\lim_{n\to\infty}(a_n) = -\infty$.

(iv) $a_n \to \infty$ as $n \to \infty$ or $a_n \to -\infty$ as $n \to \infty$.

Using the classification of divergent sequences, we can now prove some properties of them.

Example 5.3: Divergent Sequences

(i) Let (a_n) be the sequence given by $a_n = n^2$. Then (a_n) diverges to ∞.

Proof. Given $M > 0$, by the Archimedean property, and the fact that $\sqrt{M} \in \mathbb{R}^+$, these exists an $N \in \mathbb{N}$ such that $N > \sqrt{M}$. Hence, for all $n > N$ it follows that

$$a_n = n^2 > N^2 > M.$$

Hence, $\lim(a_n) = \infty$. □

(ii) Let (b_n) be the sequence given by $b_n = (-1)^n$. Then (b_n) diverges.

Proof. Suppose towards a contradiction that (b_n) converges to $b \in \mathbb{R}$. Let $\varepsilon = 1/2$. Then

$$|(-1)^n - b| < \frac{1}{2}$$

for all $n > N$. For each even $n > N$ it follows that

$$|1 - b| < \frac{1}{2}$$

and for each odd $n > N$ it follows that

$$|-1 - b| = |1 + b| < \frac{1}{2}.$$

Hence,

$$\begin{aligned} 2 &= |1 - b + 1 + b| \\ &\leq |1 - b| + |1 + b| \\ &< \frac{1}{2} + \frac{1}{2} \\ &= 1 \end{aligned}$$

for all $n > N$. This is a contradiction and hence (b_n) is divergent. □

Proving the divergence of sequences directly, much like convergence, can be difficult and lengthy. Fortunately, we can utilise other theorems to hopefully produce simpler and shorter proofs. As you may suspect, if a sequences converges, then it must be bounded. This is proven explicitly in the following theorem.

Theorem 5.4: Convergence and Bounded

All convergent sequences (a_n) are bounded.

Proof. Let $\lim(a_n) = a$. Since (a_n) is convergent, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < 1$$

for all $n > N$. Now, define the set

$$S = \{|a_1|, |a_2|, \ldots, |a_N|, |a + 1|, |a - 1|\}.$$

Letting $K = \max(S)$ it follows that

$$|a_n| \leq K$$

for all $n \in \mathbb{N}$. □

Using this theorem, we can prove some sequences are divergent by simply showing they are unbounded. This is due to the fact that the statement in Theorem 5.4 is equivalent to its contrapositive. An example utilising this theorem is as follows.

Example 5.5: Unbounded means Divergent

Let (a_n) be the sequence given by $a_n = (-1)^n n$. Then a_n is divergent.

Proof. By Example 3.3 (iii), the sequence is unbounded. Hence, by Theorem 5.4 (b_n) is divergent. □

It is important to note that not all divergent sequences are unbounded. This was shown in Example 5.3 (ii). This sequence was bounded but divergent. So proving a sequence is bounded neither proves that the sequence is convergent or divergent. In other words, boundedness is a necessary condition for convergence but not sufficient.

6 The Algebraic and Order Limit Theorems

So far, we have constructed a rigorous definition for the convergence of a sequence. An important point to note is that this definition was not constructed so as to have a tool to prove the convergence of sequences such as

$$a_n = \frac{1}{n^2}.$$

Indeed, this definition can be used to do so, and we have even put it to use in some examples. But, that is not the primary purpose for introducing it. After all, one could argue intuitively what the sequence converges to without the need for such a precise notion of convergence. This again mirrors the historical development of convergence in which the formal definition was written down by Cauchy some 150 years after the conception of calculus. So why then do we need such a rigorous definition? The purpose for this is so we can confidently prove statements about convergent sequences in general. Ultimately, it is our aim to resolve arguments about what is and is not true regarding the behaviour of limits with respect to the mathematical manipulations we intend to inflict on them. As we saw in the early chapters, the assumption of familiar properties of commutativity of addition on infinite series led to paradoxical results. These results emphasise the point that intuitive notions can only take us so far and instead we must take great care in drawing conclusions about the nature of infinite objects. To this end, we want to see how convergent sequences behave with respect to the algebraic operations of addition and multiplication. For example, do the sum of two convergent sequences converge or can they somehow diverge. Fortunately, the follow theorem demonstrates that the algebraic operations on $\mathbb{R}$ behave nicely with convergent sequences and do not lead to any nasty results.

Theorem 6.1: Algebraic Limit Theorem

Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$. Then it follows that

(i) $\lim(ca_n) = ca$ for all $c \in \mathbb{R}$.

(ii) $\lim(a_n + b_n) = a + b$.

(iii) $\lim(a_n b_n) = ab$.

(iv) $\lim(a_n/b_n) = a/b$ providing $b \neq 0$ and $b_n \neq 0$ for each $n \in \mathbb{N}$.

Proof. (i) Since $\lim(a_n) = a$, it follows for every $\varepsilon > 0$ and $c \in \mathbb{R} \setminus \{0\}$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon/|c|$$

for every $n > N$. Hence,

$$\begin{aligned} |ca_n - ca| &= |c||a_n - a| \\ &< |c|\frac{\varepsilon}{|c|} \\ &= \varepsilon. \end{aligned}$$

In the case when $c = 0$, this reduces to showing that the *constant sequence* $(0, 0, 0, \dots)$ converges to 0. This will be an exercise. □

Proof. (ii) For every $\varepsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2}$$

for every $n > N_1$. Additionally, there exists an $N_2 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{\varepsilon}{2}$$

for every $n > N_2$. Define $N = \max(N_1, N_2)$. It follows that for every $n > N$, we have

$$\begin{aligned} |(a_n + b_n) - (a + b)| &\leq |a_n - a| + |b_n - b| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

Proof. (iii) For every $\varepsilon > 0$, there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2|b| + 1}$$

for every $n > N_1$. Additionally, since (a_n) is convergent, it follows by Theorem 5.4 that it is bounded. Hence, there exists an $M \in \mathbb{R} \setminus \{0\}$ such that $|a_n| < M$ for every $n \in \mathbb{N}$. Now, as (b_n) is convergent, there exists an $N_2 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{\varepsilon}{2M}$$

for every $n > N_2$. Define $N = \max(N_1, N_2)$. Then it follows that for every $n > N$, we have

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n b_n - a_n b| + |a_n b - ab| \\ &= |a_n||b_n - b| + |b||a_n - a| \\ &< M\frac{\varepsilon}{2M} + |b|\frac{\varepsilon}{2|b| + 1} \\ &< \varepsilon. \end{aligned}$$

□

Proof. (iv) This will follow from (iii) if $\lim(1/b_n) = 1/b$. As $\lim(b_n) = b \neq 0$ it follows that there exists an $N_1 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{|b|}{2}$$

for every $n > N_1$. If follows that $|b_n| > |b|/2$ and thus

$$\frac{1}{|b_n|} < \frac{2}{|b|}$$

for all $n > N_1$. Additionally, for every $\varepsilon > 0$, there exists an $N_2 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{\varepsilon |b|^2}{2}$$

for every $n > N_2$. Define $N = \max(N_1, N_2)$. Then it follows that for every $n > N$, we have

$$\begin{aligned} \left|\frac{1}{b_n} - \frac{1}{b}\right| &= \left|\frac{b - b_n}{b_n b}\right| \\ &= \frac{|b_n - b|}{|b_n||b|} \\ &< \frac{2\varepsilon|b|^2}{2|b||b|} \\ &= \varepsilon. \end{aligned}$$

□

Whilst the algebraic limit theorem tells us that convergent sequences behave nicely with the algebraic operations of addition and multiplication, and this in turn will allow us to prove other important results, the theorem also allows us to be able to calculate the limit of more complicated sequences. More specifically, for any sequence that is constructed from the addition or multiplication of known convergent sequences, we are able to use the algebraic limit theorem to find the limit of this complex composite sequence. The following example demonstrate this point explicitly.

Example 6.2: Algebraic Limit Theorem

(i) Let (a_n) be the sequence given by $a_n = 1/n$ and (b_n) be the sequence given by $b_n = 4/n^2$. Then since $\lim(a_n) = 0$ and $\lim(b_n) = 0$, it follows that

$$\lim\left(\frac{1}{n} + \frac{4}{n^2}\right) = 0 + 0.$$

(ii) Let (a_n) be the sequence given by

$$a_n = \frac{3n^2 - 6n + 2}{5n^2 - 2n + 3}.$$

Dividing top and bottom of a_n by n^2 gives

$$a_n = \frac{3 - \frac{6}{n} + \frac{2}{n^2}}{5 - \frac{2}{n} + \frac{3}{n^2}}.$$

Now, since $\lim(k/n^\alpha) = 0$ for all $k \in \mathbb{R}$ and $\alpha \in \mathbb{N}$, it follows by Theorem 6.1 (i) and (ii) that

$$\lim\left(3 - \frac{6}{n} + \frac{2}{n^2}\right) = 3$$
$$\lim\left(5 - \frac{2}{n} + \frac{3}{n^2}\right) = 5.$$

Now by Theorem 6.1 (iv), we find that

$$\lim(a_n) = \frac{3}{5}.$$

Having verified that convergent sequences behave well with the algebraic operations on $\mathbb{R}$, the question remains as to whether convergent sequences also behave appropriately with respect to the order operation. Again, like the algebraic operations, nothing too controversial happens here either.

Theorem 6.3: Order Limit Theorem

Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$. Then,

(i) If $a_n \geq 0$ for each $n \in \mathbb{N}$, it follows that $a \geq 0$.

(ii) If $a_n \leq b_n$ for each $n \in \mathbb{N}$, it follows that $a \leq b$.

(iii) If there exists a $c \in \mathbb{R}$ satisfying $c \leq b_n$ for all $n \in \mathbb{N}$, then $c \leq b$. Similarly, if $a_n \leq c$ for each $n \in \mathbb{N}$, it follows that $a \leq c$.

Proof. (i) Assume towards a contradiction that $a < 0$. Then there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < |a|$$

for every $n > N$. Hence,

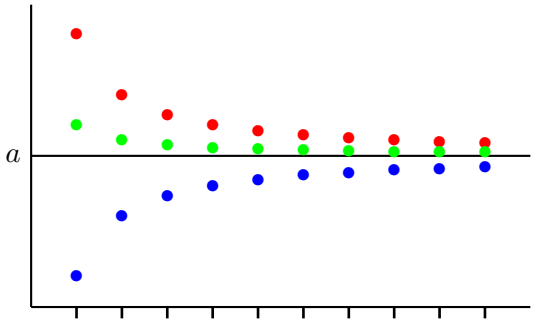
$$|a_{N+1} - a| < |a|$$
$$\Rightarrow a_{N+1} < 0.$$

This is a contradiction, and hence $a \geq 0$. □

Proof. (ii) Exercise □

Proof. (iii) Exercise □

Whilst Theorem 6.3 (iii) explores the behaviour of convergent sequences bounded from below and above separately, it doesn't address the behaviour of sequences bounded both from above and below by convergent sequences. That is, we wonder how a sequence behaves if it is eventually sandwiched between two other convergent sequences. Pictorially, we can represent this as follows:



In this picture, we have $a_n \leq b_n \leq c_n$ with a_n in red, b_n in green and c_n in blue. What this picture appears to be showing is that if b_n is eventually sandwiched between two other convergent sequences, then b_n is convergent. This is true in general and is proven by the following theorem.

Theorem 6.4: Squeeze Theorem

If $a_n \leq b_n \leq c_n$ for all $n > m$ and $\lim(a_n) = \lim(c_n) = a$, then $\lim(b_n) = a$.

Proof. For every $\varepsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for every $n > N_1$. Similarly, there exists an $N_2 \in \mathbb{N}$ such that

$$|c_n - a| < \varepsilon$$

for every $n > N_2$. Define $N = \max(N_1, N_2, m)$. Then it follows that for every $n > N$, we have

$$a - \varepsilon < a_n \leq b_n \leq c_n < a + \varepsilon$$
$$\Rightarrow |b_n - a| < \varepsilon.$$

Hence $\lim(b_n) = a$. □

This theorem can be utilised to prove the convergence of even more complicated sequences. By finding pairs of convergent sequences that converge to the same value and eventually bound the complex sequence from above and below, we are able to not only able to deduce that the sequence is convergent but also what it converges to. Some examples of this are as follows.

Example 6.5: Squeeze Theorem

(i) Let (a_n) be the sequence given by

$$a_n = \frac{\sin(n)}{n}.$$

Since $-1 \leq \sin(n) \leq 1$ for every $n \in \mathbb{N}$, it follows that

$$-\frac{1}{n} \leq a_n \leq \frac{1}{n}.$$

for all $n \in \mathbb{N}$. Using $\lim(-1/n) = \lim(1/n) = 0$, it follows by the squeeze theorem that $\lim(a_n) = 0$.

(ii) Let (a_n) be a sequence given by $a_n = \dfrac{x^n}{n!}$ for fixed positive $x \in \mathbb{R}^+$. Since $n!, x^n > 0$ for each $n \in \mathbb{N}$, it follows that

$$0 \leq \frac{x^n}{n!}.$$

Let $m = \min(\{n \in \mathbb{N} : n \geq x\})$. Then for $n > m$, we have that

$$\frac{x^n}{n!} = \frac{x}{1}\frac{x}{2}\cdots\frac{x \cdot x \cdots x}{m(m+1)\cdots(n-1)}\frac{x}{n} \leq \frac{x}{1}\frac{x}{2}\cdots\frac{x}{m-1}\frac{x}{n}.$$

Defining

$$c = \frac{x}{1}\frac{x}{2}\cdots\frac{x}{m-1}$$

we have that

$$0 \leq a_n \leq \frac{cx}{n}$$

for all $n > m$. Since $\lim(cx/n) = 0$, it follows by the squeeze theorem that $\lim(a_n) = 0$.

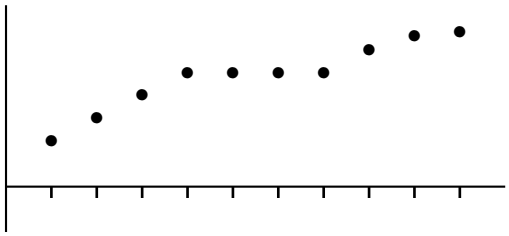
7 Monotone Sequences

For any convergent sequence (a_n), we have so far only been able to prove it converges if the value of the limit is known in advance. For example, to prove $a_n = 1/n$ converges to zero we necessarily needed to know, or more accurately guess, that $\lim(1/n) = 0$. From here, we were able to prove its convergence. However, there are many sequences, particularly recursive ones, that are convergent but it isn't at

all clear what they converge to. For example, take the sequence

$$a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}$$

with $a_1 = 1$. This sequence does in fact converge, but how can we prove this? Ideally, we want to be able to prove some properties of the sequence (a_n) that ensure its convergence. But, we only have a necessary property for convergence, namely that the sequence must be bounded. It isn't too hard to come up with a sequence that is bounded but not convergent, hence this condition is not sufficient for convergence. What additional property must we then add to the sequence so as to ensure its convergence? Well, if the sequence is say bounded above, it seems that if the sequence was always increasing, it must converge. Intuitively, if the sequence is both bounded from above and always increasing, it must be getting closer and closer to some value, and that value should be the least upper bound for the set containing all the terms of the sequence. In a similar way, if the sequence was bounded from below and decreasing it seems that it would also have to converge. To prove this suspicion, we must first define what we mean by an *increasing* or *decreasing* sequence. Like with most definitions, drawing a picture of what we expect such a sequence to look like will help.



This picture represents what we expect an increasing sequence to look like. A similar one could be drawn for a decreasing sequence. With these in mind, we arrive at our definition.

Definition 7.1: Monotone Sequence

A sequence (a_n) is said to be ***monotone increasing*** if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$.

Similarly, a sequence (b_n) is said to be ***monotone decreasing*** if $b_n \geq b_{n+1}$ for all $n \in \mathbb{N}$

If a sequence is either monotone increasing or monotone decreasing, then it is said to be ***monotone.***

This definition allows us to identify if a sequence is monotone or not. To do this, it is common to consider the difference of two subsequent terms of the sequence. Depending on the sign of this difference, this determines if the sequence is monotone decreasing or increasing. This is outlined explicitly in the following examples.

Example 7.2: Monotone Sequence

(i) Let (a_n) be the sequence given by $a_n = \sqrt{n}$. Since

$$\begin{aligned}\sqrt{n+1} - \sqrt{n} &= (\sqrt{n+1} - \sqrt{n})\frac{(\sqrt{n+1} + \sqrt{n})}{(\sqrt{n+1} + \sqrt{n})} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}} \\ &> 0.\end{aligned}$$

Hence $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$ and thus (a_n) is monotone increasing.

(ii) Let (b_n) be the sequence given by $b_n = 1/n$. Then

$$b_n - b_{n+1} = \frac{(n+1) - n}{n(n+1)} = \frac{1}{n(n+1)} \geq 0.$$

Hence (b_n) is monotone decreasing.

Recall that our primary motivation for introducing monotone sequences was to show that both bounded and monotone sequences were necessarily convergent. If this were the case, we would be able to deduce the convergence of sequences by showing they satisfy these properties without any knowledge of the numbers they converge to. To this end, we arrive at the following theorem.

Theorem 7.3: Monotone Convergence Theorem

Every bounded monotone sequence is convergent.

Proof. Let (a_n) be bounded and monotone. Then for every $n \in \mathbb{N}$, we have either that $a_n \leq a_{n+1}$ or $a_n \geq a_{n+1}$ as well as $|a_n| < M$ for some $M \in \mathbb{R}$. Define the set

$$A = \{a_n : n \in \mathbb{N}\}.$$

Since A is bounded, by the completeness axiom both the supremum and infimum exists. Let $\alpha = \sup(A)$ and $\beta = \inf(A)$. Then it follows that for every $\varepsilon > 0$ there exists $N_1 \in \mathbb{N}$ and $N_2 \in \mathbb{N}$ such that

$$\alpha - \varepsilon < a_{N_1}$$

and

$$a_{N_2} < \beta + \varepsilon.$$

Now if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$ it follows that $a_{N_1} < a_n$ for all $n > N_1$. Hence for every $n > N_1$, it would follow that

$$\begin{aligned}&\alpha - \varepsilon < a_n \leq \alpha < \alpha + \varepsilon \\ &\Rightarrow |a_n - \alpha| < \varepsilon.\end{aligned}$$

In the other case when $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$, it follows that $a_n < a_{N_2}$ for

all $n > N_2$. Hence, for every $n > N_2$, it would follow that

$$\begin{aligned} &\beta - \varepsilon < \beta \leq a_n < \beta + \varepsilon \\ &\Rightarrow |a_n - \beta| < \varepsilon. \end{aligned}$$

□

So it is the case that our suspicion of bounded monotone sequences must be convergent is true. Let's see if we can utilise this now to prove the convergence of a sequence without any knowledge of the value it converges to.

Example 7.4: Monotone Convergence Theorem

Let (b_n) be the sequence given by

$$b_{n+1} = \frac{1}{4 - b_n}$$

and $b_1 = 3$. Now $0 < b_1 \leq 3$. Then let $0 < b_n \leq 3$ for some $n \in \mathbb{N}$. It follows that

$$1/4 < 1/(4 - b_n) \leq 1$$

and hence $0 < b_{n+1} \leq 3$. By induction, $0 < b_n \leq 3$ for all $n \in \mathbb{N}$. Similarly, assume $b_{n+1} \leq b_n$ for some $n \in \mathbb{N}$. Then

$$\begin{aligned} &4 - b_n \leq 4 - b_{n+1} \\ &\Rightarrow \frac{1}{4 - b_n} \geq \frac{1}{4 - b_{n+1}} \\ &\Rightarrow b_{n+1} \geq b_{n+2}. \end{aligned}$$

Since $b_1 = 3 \geq 1 = b_2$, it follows by induction that $b_{n+1} \leq b_n$ for every $n \in \mathbb{N}$. That is, (b_n) is monotone decreasing. By the monotone convergence theorem, it follows that (b_n) is convergent. Let $\lim(b_n) = b$. It follows that $\lim(b_{n+1}) = b$ and

$$\begin{aligned} b &= \frac{1}{4 - b} \\ &\Rightarrow b^2 - 4b + 1 = 0 \\ &\Rightarrow b = 2 \pm \sqrt{3}. \end{aligned}$$

Since $b_n \leq 3$ for all $n \in \mathbb{N}$, it must be that $b = 2 - \sqrt{3}$.

It is worth commenting that this theorem does not tell us that all convergent sequences are monotone and bounded. Whilst it does allow us to prove the convergence of some convergent sequences without the knowledge of the exact limit, it does not allow us to prove the convergence of all of them. This fundamentally comes down to the fact that a sequence being monotone is not necessary for it to be convergent.

It may have dawned on you that there is an obvious relationship between bounded monotone sequences and suprema/infima. In particular, if (a_n) is a monotone increasing sequence that is bounded above, the value it converges to seems to be

$\sup(S)$ where

$$S = \{a_n : n \in \mathbb{N}\}.$$

It appears that there is a relationship between the supremum of a set and the existence of a convergent sequence (a_n). This is indeed the case and is given precisely by the following theorem.

Theorem 7.5: Sequences and Suprema

Let $S \subset \mathbb{R}$ be a non-empty subset bounded above and let $\alpha \in \mathbb{R}$ be an upper bound for S. Then $\alpha = \sup(A)$ if and only if there exists a sequence (a_n), with $a_n \in S$ for each $n \in \mathbb{N}$, such that $\lim(a_n) = \alpha$.

Proof. ($\Rightarrow$) Let $\alpha = \sup(A)$. Then for every $n \in \mathbb{N}$, it follows that there exists an $a_n \in S$ such that

$$\alpha - \frac{1}{n} \leq a_n \leq \alpha.$$

This defines a sequence (a_n) and since $\lim(1/n) = 0$, it follows by Theorem 6.4 that $\lim(a_n) = \alpha$.
($\Leftarrow$) Suppose there exists a sequence (a_n) such that $a_n \in A$ for each $n \in \mathbb{N}$ and $\lim(a_n) = \alpha$. Then for every $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ with

$$\begin{aligned} &|a_N - \alpha| < \varepsilon \\ &\Rightarrow -\varepsilon < a_N - \alpha \\ &\Rightarrow \alpha - \varepsilon < a_N. \end{aligned}$$

Hence by Theorem 4.2 in Chapter 1, it follows that $\sup(A) = \alpha$. □

As one might expect, there is a similar version of this theorem for infima, the proof of which will be an exercise.

Now, this theorem can now be utilised to prove the suprema of given sets more easily. That is, by simply showing there exists a convergent sequence in the set that converges to an upper bound of the set, then that upper bound must be the supremum. An explicit proof of this type is given in the following example.

Example 7.6: Sequence and Suprema

Let $A = \{1 - 1/n : n \in \mathbb{N}\}$. Clearly $0 \in A$ and $1 > 1 - 1/n$ for every $n \in \mathbb{N}$. Thus 1 is an upper bound for A and A is non-empty. Let (a_n) be the sequence $a_n = 1 - 1/n$. Then each $a_n \in A$, by the definition of A. Since $\lim(a_n) = 1$ it follows by Theorem 7.5 that $\sup(A) = 1$.

8 Subsequences and the Bolzano-Weierstrass Theorem

Analysing mathematical structures as a whole can often be complicated. Instead, by analysing the behaviour of the component pieces, we may be able to extrapolate

some knowledge about the entire structure. In this case, the structures being studied are sequences, and their component pieces are *subsequences*. This is precisely what you expect. That is, sequences within the larger sequence. For example, take the sequence

$$(1, 2, 3, 4, \dots).$$

Then a subsequence of this could be $(2, 4, 6, \dots)$ or $(1, 3, 5, \dots)$. By analysing the behaviour of these subsequences, perhaps we can deduce something about the behaviour of the sequence. With these intuitive ideas in hand, we now arrive at the formal definition.

Definition 8.1: Subsequence

Let (n_k) be an increasing sequence of positive integers. That is,

$$n_1 < n_2 < n_3 < \dots$$

for each $n_k \in \mathbb{N}$. Then, given a sequence (a_n), the sequence given by (a_{n_k}) is called a ***subsequence*** of (a_n).

Whilst the opening introduced a few examples of subsequences, it wasn't done using this new notation. Some explicit examples of subsequences with this notation are given as follows.

Example 8.2: Subsequence

(i) Let (a_n) be the sequence given by $a_n = (-1)^n$. Defining $n_k = 2k$ with $k \in \mathbb{N}$, it follows that (n_k) is a sequence of positive integers given explicitly by

$$(n_k) = (2, 4, 6, \dots).$$

Then the subsequence (a_{n_k}) is given by

$$(a_{n_k}) = (1, 1, 1, \dots).$$

(ii) Let (b_n) be the sequence given by $b_n = 1/n$. Then the sequence given by

$$(n_k) = (10, 100, 1000, \dots)$$

gives the subsequence

$$(b_{n_k}) = \left(\frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \dots\right).$$

With the definition of subsequences, we would like to see if these is some connection between the behaviour of convergent sequences and their corresponding subsequences. Intuitively, it would seem to be the case that if a sequence was convergent, then all its subsequences are also convergent. This is exactly the case and is proven in the following theorem.

Theorem 8.3: Convergence and Subsequences

A sequence (a_n) converges to a if and only if every subsequence of (a_n) converges to a.

Proof. ($\Rightarrow$). Suppose that (a_n) converges to a and let (a_{n_k}) be a subsequence of (a_n). Then for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for every $n > N$. Since $n_k \geq k$ for each $k \in \mathbb{N}$, it follows that for every $k > N$, we have

$$|a_{n_k} - a| < \varepsilon.$$

($\Leftarrow$) *Exercise* □

Whereas this theorem is not that helpful for identifying if a sequence is convergent, since it requires one to check every subsequence is convergent, it is useful for proving a sequence is divergent. By taking the contrapositive of Theorem 8.3, we are able to state the following useful corollary.

Corollary 8.4: Divergence and Subsequences

If there exists two distinct subsequences of a sequence (a_n) that converge to different limits, then (a_n) diverges.

With this, we can now prove the divergence of a given sequence by simply showing that there exists two subsequences converging to different values. This is often far simpler than the proofs that were constructed previously. An explicit example of a proof of this type is given by the following.

Example 8.5: Convergence and Subsequences

Let (a_n) be the sequence given by $a_n = (-1)^n$. Defining $n_k = 2k$ and $\tilde{n}_k = 2k - 1$, with $k \in \mathbb{N}$, gives the subsequences

$$(a_{n_k}) = (1, 1, 1, \dots)$$

and

$$(a_{\tilde{n}_k}) = (-1, -1, -1, \dots).$$

Since $\lim(a_{n_k}) = 1$ and $\lim(a_{\tilde{n}_k}) = -1$ it follows by Corollary 8.4 that (a_n) is divergent.

We now ponder the question, given an arbitrary sequence, does there always exist a convergent subsequence? After a little thought, it is fairly clear that not all sequences contain a convergent subsequence. A example of this could be $(1, 2, 3, \dots)$. Since this sequence diverges to infinity, all its subsequences also diverge to infinity. So whilst not all sequences contain convergent subsequences, maybe they do contain subsequences that satisfy a weaker condition, namely being monotone. This

is exactly the case and is proven in the following theorem.

Theorem 8.6: Monotone Subsequence

Every sequence has a monotone subsequence.

Proof. We say a_n is a *peak* if $a_n \geq a_m$ for all $m > n$. It must be that there are either infinitely many peaks or finitely many. Assume first that there are finitely many peaks. Then there exists an $N \in \mathbb{N}$ in which $a_N \geq a_m$ for all $m > N$. That is, a_N is the last peak or smallest peak. Letting $n_1 = N + 1$, it follows then since a_N was the last peak, it must be that a_{n_1} is not a peak and thus there exists an $n_2 \in \mathbb{N}$ with $n_2 > n_1$ and $a_{n_2} > a_{n_1}$. Similarly, a_{n_2} cannot be a peak, and hence there exists an $n_3 > n_2$ such that $a_{n_3} > a_{n_2}$. Continuing this constructs a monotone increasing subsequence (a_{n_k}).
If there are infinitely many peaks then let a_{n_k} denote the k-th peak. It follows that $a_{n_k} \geq a_{n_{k+1}}$ and hence (a_{n_k}) is a monotone decreasing subsequence. □

Recall that we first started by considering the question, do all sequences contain a convergent subsequence. This was unfortunately not the case, but we now wonder that if we restrict ourselves to a certain class of sequences, it maybe true. Given we already know that all bounded monotone sequences are convergent and every sequence contains a monotone subsequence, the condition we need to add is boundedness. That is, piecing all these theorems together, we seem to conclude that every bounded sequence should contain a convergent subsequence. This statement is a famous theorem known as the *Bolzano-Weierstrauss theorem* and is stated as follows.

Theorem 8.7: Bolzano-Weierstrauss Theorem

Every bounded sequence has a convergent subsequence.

Proof. If (a_n) is a bounded sequence, it follows by Theorem 8.6 that there exists a monotone subsequence (a_{n_k}). Also, since (a_n) is bounded, so is (a_{n_k}). By the Theorem 7.3 it follows that (a_{n_k}) is convergent. □

9 The Cauchy Criterion

It was commented earlier that a sequence being monotone was not a necessary condition for convergence. We still then wonder if there is a necessary and sufficient condition for a sequence to be convergent. This would be particularly useful if we were wishing to prove the convergence of a sequence that was not monotone and we didn't know what it converged to. One property that may seem rather reasonable for convergence is that the difference between subsequent terms of a sequence gets arbitrarily small. That is, for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ so that $|a_{n+1} - a_n| < \varepsilon$ for all $n > N$. Intuitively, this seems like a reasonable property since if a sequence is convergent, eventually the 'jumps' between each term of the sequence to the next should get smaller and smaller. One then may conjecture that this is a necessary

and sufficient condition for a sequence to be convergent. Unfortunately, this is not the case and a counterexample is given by the sequence (a_n) with $a_n = \sqrt{n}$. A short calculation shows that

$$|a_{n+1} - a_n| < \frac{1}{\sqrt{n}}.$$

Hence, if $N = \min(\{n \in \mathbb{N} : 1 < n\varepsilon^2\})$, for every $n > N$ it follows that

$$|a_{n+1} - a_n| < \varepsilon.$$

So the sequence (a_n) has the subsequent terms getting arbitrarily close, but clearly (a_n) is not convergent. So what went wrong and our we even on the right track? Well, even though the terms next to one another got arbitrarily close, this is not necessarily the case for terms further apart. That is, we deduced that $|a_{n+1} - a_n|$ is small, but we know nothing about $|a_{n+500} - a_n|$. So perhaps the fix is to say that eventually the difference between all terms of the sequence get arbitrarily small. This is precisely the condition we need and is known as the *Cauchy criterion*. Sequences that satisfy such properties are known as *Cauchy sequences* and are defined as follows.

Definition 9.1: Cauchy Sequence

A sequence (a_n) is a ***Cauchy sequence*** if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon$$

for all $n, m > N$.

Since our aim here is to show that the *Cauchy criterion* is both necessary and sufficient for convergence, we should first show that Cauchy sequences satisfy the same properties as convergent ones, namely that they are bounded. This is proven in the following theorem.

Theorem 9.2: Cauchy sequences are bounded

All Cauchy sequences are bounded.

Proof. There exists an $N \in \mathbb{N}$ such that

$$|a_n - a_m| < 1$$

for all $n, m > N$. Let S be the set

$$S = \{|a_1|, |a_2|, \ldots, |a_N|, |a_{N+1}| + 1\}$$

and define $b = \max(S)$. If follows that for all $m > N$, we have that

$$\begin{aligned} |a_m| &= |a_m - a_{N+1} + a_{N+1}| \\ &\leq |a_m - a_{N+1}| + |a_{N+1}| \\ &< 1 + |a_{N+1}| \\ &\leq b \end{aligned}$$

Similarly, $|a_m| \leq b$ for all $m \leq N$. By Theorem 3.2, it follows that (a_n) is bounded. □

So it is the case that like convergent sequences, all Cauchy sequences are bounded. We now want to prove the result we suspected from the outset, and was the reason for introducing Cauchy sequences, namely, all Cauchy sequences are convergent.

Theorem 9.3: Cauchy criterion for convergence

A sequence is convergent if and only if it is Cauchy.

Proof. ($\Rightarrow$) Suppose $\lim(a_n) = a$. Hence, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2}$$

for all $n > N$. Then for every $n, m > N$, it follows that

$$\begin{aligned} |a_n - a_m| &= |a_n - a + a - a_m| \\ &\leq |a_n - a| + |a - a_m| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

So (a_n) is Cauchy.
($\Leftarrow$) Suppose (a_n) is Cauchy. By Theorem 9.2, it follows that (a_n) is bounded and by Theorem 8.7 there exists a convergent subsequence (a_{n_k}). Say $\lim(a_{n_k}) = a$. Now, for every $\varepsilon > 0$, there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a_m| < \frac{\varepsilon}{2}$$

for all $n, m > N_1$. Also, there exists an $N_2 \in \mathbb{N}$ such that

$$|a_{n_k} - a| < \frac{\varepsilon}{2}$$

for all $k > N_2$. Define $N = \max(N_1, N_2)$. Since $n_k \geq k$ and thus $n_{k+1} > k$,

it follows that for every $k > N$ we have

$$\begin{aligned} |a_k - a| &= |a_k - a_{n_{k+1}} + a_{n_{k+1}} - a| \\ &\leq |a_k - a_{n_{k+1}}| + |a_{n_{k+1}} - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence (a_n) converges. □

Finally, we have a condition on a sequence for it to be convergent. Unlike the monotone case, the Cauchy criterion must hold for every convergent sequence. Hence, we can prove that a given sequence is convergent by showing it is Cauchy. The advantages of this are that we do not need to know the value the sequence converges to, like in the monotone case, but additionally we can also analyse the convergence of non-monotone sequences. An example of this type is given as follows.

Example 9.4: Cauchy criterion for convergence

Let (a_n) be a sequence given by

$$a_{n+1} = 1 + \frac{1}{a_n}$$

and $a_1 = 1$. Then (a_n) is Cauchy and hence convergent.

Proof. Since $a_n \geq 1$ for every $n \in \mathbb{N}$ it follows that

$$a_n a_{n-1} = 1 + a_{n-1} \geq 2.$$

Hence, for every $n > 1$, it follows that

$$\begin{aligned} |a_{n+1} - a_n| &= \left| \frac{1}{a_n} - \frac{1}{a_{n-1}} \right| \\ &= \left| \frac{a_{n-1} - a_n}{a_n a_{n-1}} \right| \\ &\leq \frac{1}{2} |a_n - a_{n-1}| \\ &\leq \frac{1}{2^{n-1}} |a_2 - a_1|. \\ &= \frac{1}{2^{n-1}}. \end{aligned}$$

Using this inequality, for $k \geq 1$ and let $n > 2$, we have that

$$\begin{aligned}
|a_{n+k} - a_n| &\leq |a_{n+k} - a_{n+k-1}| + |a_{n+k-1} - a_{n+k-2}| + \cdots + |a_{n+1} - a_n| \\
&\leq \frac{1}{2^{n+k-2}} + \frac{1}{2^{n+k-3}} + \cdots + \frac{1}{2^{n-1}} \\
&= \left(\frac{1}{2}\right)^{n-1} \left(\frac{1 - \left(\frac{1}{2}\right)^k}{1 - \frac{1}{2}}\right) \\
&< \frac{1}{2^{n-2}}
\end{aligned}$$

Now, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$, with $N \geq 2$, such that

$$\frac{1}{2^{N-2}} < \varepsilon.$$

Hence, for every $n, m > N$ we have that

$$|a_n - a_m| < \varepsilon.$$

Since (a_n) is Cauchy, it is convergent. Let $\lim(a_n) = \alpha$. Then by the algebraic property of limits, and the ordering properties, it follows that

$$\begin{aligned}
&\alpha = 1 + \frac{1}{\alpha} \\
&\Rightarrow \alpha^2 - \alpha - 1 = 0 \\
&\Rightarrow \alpha = \frac{1 + \sqrt{5}}{2}.
\end{aligned}$$

The number α is known as *The Golden Ratio.* □

This concludes the chapter on sequences. The primary motivation for exploring this subject is so that we could analyse the behaviour of infinite series. We will see explicitly in the next chapter how the framework we have constructed here can be used to great effect to explore the properties of infinite series. Further, we will see how the properties of sequences can also be utilised to analyse the properties of functions. It is therefore crucial that you understand all the theorems and definitions in the chapter since they will often be referenced in future chapters.